DT081A - Signal and Image Processing
Exercise 4 Solutions

Question 1. Write the definitions of a signal's:

i) Frequency Spectrum

ii) Amplitude spectrum

iii) Phase spectrum

Answer:

i) The frequency spectrum is defined via

$$\mathbb{R} \to \mathbb{C} \tag{1}$$
$$\omega \mapsto X(\omega) = \langle x(t), e^{j\omega t} \rangle \tag{2}$$

ii) Amplitude spectrum is defined via

$$\mathbb{R} \to \mathbb{R}_+ \tag{3}$$
$$\omega \mapsto \|X(\omega)\|_2 \tag{4}$$

iii) The phase spectrum is defined as

$$\mathbb{R} \to (-\pi, \pi] \tag{5}$$
$$\omega \mapsto \arctan\left(\frac{\mathrm{Im}(X(\omega))}{\mathrm{Re}(X(\omega))}\right) + \frac{\pi}{2}(1 - \mathrm{sign}(\mathrm{Re}(X(\omega)))) \tag{6}$$

Question 2. What is the principle of these spectra? Why is it useful in signal processing?

Answer: The principle of these spectra is to decompose a signal into its constituent frequency components. This is useful in signal processing because it allows us to:

- Analyze the frequency content of signals
- Design and implement filters
- Compress signals by removing less important frequency components
- Identify and remove noise from signals
- Perform frequency-domain analysis of systems

Question 3. In this course, we mainly (only) consider LTI systems.

1. What does the abbreviation LTI stand for, and why is it useful to consider such systems?
2. Formulate mathematically what LTI-systems imply.

Answer:

- LTI stands for Linear Time-Invariant. These systems are useful to consider becauset they are mathematically tractable, many real-world systems can be approximated as LTI and they have well-defined properties that simplify analysis and theory.
- Mathematically, LTI systems imply linearity, meaning an LTI system H has to satisfy linearity: $H\{ax_1(t) + bx_2(t)\} = aH\{x_1(t)\} + bH\{x_2(t)\}$. They also need to be time-invariant, meaning if $y(n) = H\{x(n)\}$, then $y(n - n_0) = H\{x(n - n_0)\}$.

Where H is the system operator, x(t) is the input, and y(t) is the output.

Question 4. Write the mathematical expression of the impulse response of a causal system.

Answer: For a causal system, the impulse response $h(n)$ needs to be of the form

$$h(n) = \begin{cases} 0 & n < 0 \\ h^+(n) & n \geq 0. \end{cases}$$

Question 5. Give three ways an LTI is defined, both time-continuous and time-discrete LTI systems.

Answer: Continuous/discrete LTI systems are fully defined by any of (Lecture 6 slide 10)

- Differential equation/difference equation
- Impulse response
- Its transform H(s) (e.g. Fourier)

Question 6. How can the output $y(t)$ of a time-discrete LTI system be computed (several options)?

Answer: The output of a time-discrete LTI system can be computed using

- the convolution sum $y(t) = \sum_{k=-\infty}^{\infty} h(t)x(t-k)$,
- the difference equation,
- or frequency domain multiplication.

Question 7.

1. Write the relationship between the input and output frequency spectra, and the system frequency response, both for a time-continuous and time-discrete LTI system.
2. Derive from this relationship the following relationships: i) amplitude spectra of input and output; ii) phase spectra of input and output.

Answer: These answers can be found in lecture 6 , slides 10 ++

1. For time-continuous LTI systems, the spectrum can be obtained via multiplication (of the Fourier transforms) $Y(\omega) = H(\omega)X(\omega)$. For time-discrete LTI systems this takes the form $Y(e^{j\Omega}) = H(e^{j\Omega})X(e^{j\Omega})$.

2. (i) The amplitude gets multiplied as

$$|Y(\omega)| = |H(\omega)| \cdot |X(\omega)|, \tag{7}$$

meaning $|H(\omega)|$ is a factor of dampening or amplification of the signal.

(ii) The phases are mapped via $\angle Y(\omega) = \angle H(\omega) + \angle X(\omega)$.

(The same relationships hold for discrete systems, replacing ω with $e^{j\Omega}$)

Question 8. Assume you have managed to identify a system's amplification and phase shift for all frequencies ω. Formulate how you compute the impulse response.

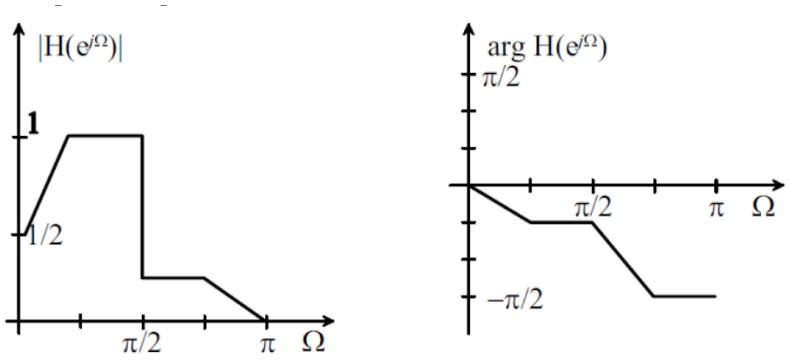
Answer: To compute the impulse response h(t) from the system's frequency response $H(\omega)$:

- Construct the frequency response (lecture 6 slide 13+): $H(\omega) = |H(\omega)|e^{j\angle H(\omega)}$
- Take the inverse Fourier transform to get the impulse response:

$$h(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} H(\omega)e^{j\omega t}d\omega$$

For discrete-time systems, use the inverse DTFT: $h(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} H(e^{j\Omega})e^{j\Omega n}d\Omega$

Question 9. A time-discrete LTI system has the following amplitude response and phase response.



1. Write with a mathematical expression how the spectrum of the input signal is related to the spectrum of the output signal.
2. What is the output signal, $y(n)$, if the input signal is $x(n) = 3 + 2\cos(n - 0.1) - 5\sin(2n + \pi/8)$
3. What new frequencies are there in the output? What is the reason?

Answer:

1. $Y(e^{j\omega}) = |H(e^{j\omega})|e^{j\angle H(e^{j\omega})}X(e^{j\omega})$

2. To find $y(n)$, we need to apply the system's frequency response to each component of $x(n)$:

$$y(n) = 3|H(0)| + 2|H(1)|\cos(n - 0.1 + \angle H(1)) - 5|H(2)|\sin(2n + \pi/8 + \angle H(2)) \tag{8}$$
$$= \frac{3}{2} + 2\cos\left(n - 0.1 - \frac{\pi}{6}\right) - \frac{5}{4}\sin\left(2n + \frac{\pi}{8} - \frac{\pi}{2}\right) \tag{9}$$
$$= \frac{3}{2} + 2\cos\left(n - 0.1 - \frac{\pi}{6}\right) - \frac{5}{4}\sin\left(2n - \frac{3\pi}{8}\right) \tag{10}$$

3. There are no new frequencies in the output. The reason is that an LTI system cannot create new frequency components; it can only modify the amplitude and phase of existing frequencies in the input signal.

Question 10. LTI systems can be analyzed and identified by letting a known signal pass the system.

1. Write the principle to identify the frequency function $H(\omega)$.
2. What characteristic of the input signal $x(t)$ is important to identify the full $H(\omega)$?
3. Mention four different input signals that give sufficient information about how the system affects sinusoids of different frequencies.
4. What are the advantages and disadvantages of using these input signals?

Answer:

1. Principle: for any ω, such taht $X(\omega), \neq 0$, we can use $H(\omega) = \frac{Y(\omega)}{X(\omega)}$, where $X(\omega)$ and $Y(\omega)$ are the Fourier transforms of the input and output signals, respectively.
2. The input signal should have non-zero amplitude for all frequencies of interest to identify the full $H(\omega)$ (i.e. $X(\omega), \neq 0$, for all ω).
3. Four input signals (see lecture 6, slides 23 ++):
   - Many sinusoids
   - Chirp (i.e. sinusoid with increasing frequency)
   - Impulse (i.e. dirac)
   - White noise
4. Advantages and disadvantages:
   - Many sinusoids: Allows for the choice of a customized frequency coverage, allowing easy interpretation and analysis of the system response to each frequency. Can be time-consuming if a large range of frequencies needs to be covered.
   - Chirp: Good frequency coverage with only a single measurement, but leads to a signal-length vs. frequency-coverage trade-off.
   - Impulse: Very simple, but must be large to give a good result. Real-world signals are never actually dirac-impulses. Furthermore, a signal that is too strong may cause damage to more sensitive systems.
   - White noise: Contains all frequencies and is easy to generate, need to wait long enough to hit the desired frequencies as they are targeted randomly.

Question 11. A time-continuous, LTI system is stimulated with the input signal

$$x(t) = 2 + 3\cos(2\pi t + 0.2) + 1.3\cos(3\pi t - 0.8) + 4\cos(\pi t)$$

whereby the following output signal is generated:

$$y(t) = 0.5 + 6\cos(2\pi t + 0.1) + 1.95\cos(3\pi t - 1.0) + 2\cos(\pi t - 0.05)$$

Plot the points in the amplification spectrum and phase shift spectrum that can be known from this.

Answer: To generate the graph, let us investigate the points in the amplification spectrum $|H(\omega)|$ and phase shift spectrum $\angle H(\omega)$:

$$|H(0)| = \frac{1}{4} \tag{11}$$
$$|H(\pi)| = \frac{1}{2} \tag{12}$$
$$\angle H(\pi) = -0.05 \tag{13}$$
$$|H(2\pi)| = 2 \tag{14}$$
$$\angle H(2\pi) = -0.1 \tag{15}$$
$$|H(3\pi)| = \frac{3}{2} \tag{16}$$
$$\angle H(3\pi) = -0.2 \tag{17}$$



Figure 1: Amplification spectrum

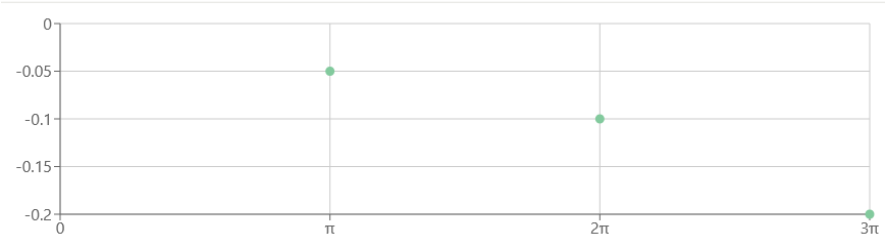


Figure 2: Phase shift spectrum

Filter Design – Spectrum

Question 12. Describe the principle of filtering.

Answer: The principle of filtering is to selectively modify or remove certain frequency components from a signal while allowing others to pass through. This is achieved by designing a system (filter) with a specific frequency response that attenuates unwanted frequencies and preserves desired frequencies.

Question 13. What is the difference between an LTI system and a filter?

Answer: An LTI system is a general concept that describes any system that is both linear and time-invariant. A filter is often a specific LTI system designed with the purpose of selectively modifying or removing certain frequency components from a signal. Thismeans, there are LTI systems whose amplification/dampening and phase-shift have not been consciously selected for any particular purpose.

Question 14. What characteristics of a filter need to be considered in its design?

Answer: Some characteristics that need to be considered are

- It being an LTI system
- Pass band
- Stop band
- The delays introduced by existing feedback connections
- Instabilities
- Phase distortions
- (Computational) complexity
- Errors introduced by quantization (for digital signals)

Question 15. What is an ideal filter? Draw the difference between an ideal filter and a "real" filter. Show in the figure what ripple is, and where it is manifested.

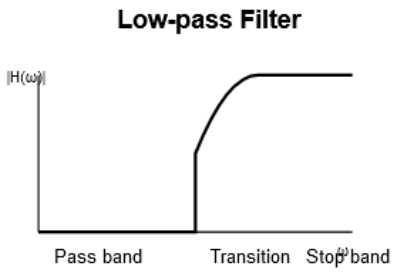
Answer: An ideal filter has a perfectly flat passband with unity gain, infinite attenuation in the stopband, and an infinitely sharp transition between passband and stopband.

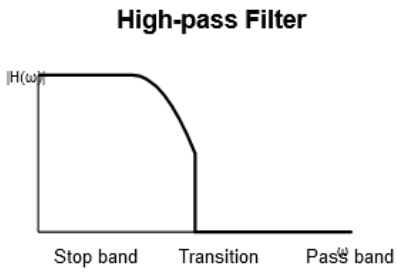


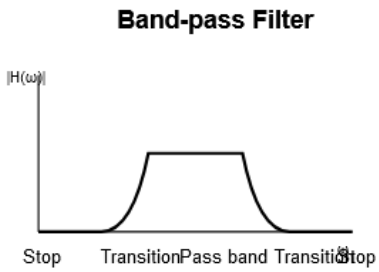
Figure 3: Example of a real filter with ripple (i.e. fluctuations in frequency in the stopband and/or passband)

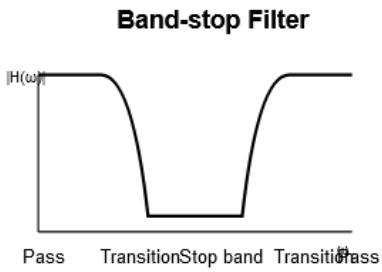
Question 16. Draw the amplification spectrum (amplitude spectrum) of a low pass, high-pass, band-pass and a band-stop filter. Indicate pass band, stop band and transition band.

Answer:









Question 17. What filter type shall you use if you want to have maximal flat amplification in the pass band and a maximal steep filter. (Search the web if you don't know.)

Answer: For maximally flat passband response and steep rolloff, you should use a Butterworth filter. Butterworth filters are known for their maximally flat magnitude response in the passband and a relatively steep rolloff.

Question 18.

1. Draw a block diagram with all the parts of a digital filter. Briefly write what each block is for task.
2. Explain the advantages and disadvantages of digital filters.

Answer:

1. Answers to this question can be found in lecture 7 slide 21+.



   - The Antialiasing filter removes frequencies above half the sampling frequency (see Nyquist-Shannon Sampling Theorem)
   - Converts analogue to digital
   - Filter desired frequencies
   - Back to continuous
   - Removes frequencies above half the sampling frequency

2. Advantages and disadvantages of digital filters:

   Advantages:

   - Arbitrarily tight tolerances
   - Almost unaffected by external factors
   - Can be reproduced exactly
   - Easy to change

   Drawbacks:

   - Only band-limited signals
   - Rounding and quantization can cause errors
   - Not too high frequencies

Question 19. Draw the structure of a general FIR and a general IIR filter.

Answer: The following figures are depicted in Lecture 5 slides 9 and 13

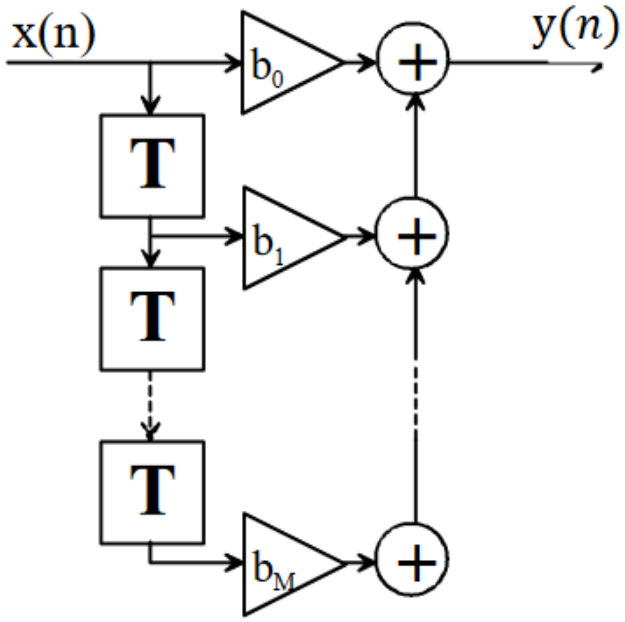


Figure 4: FIR

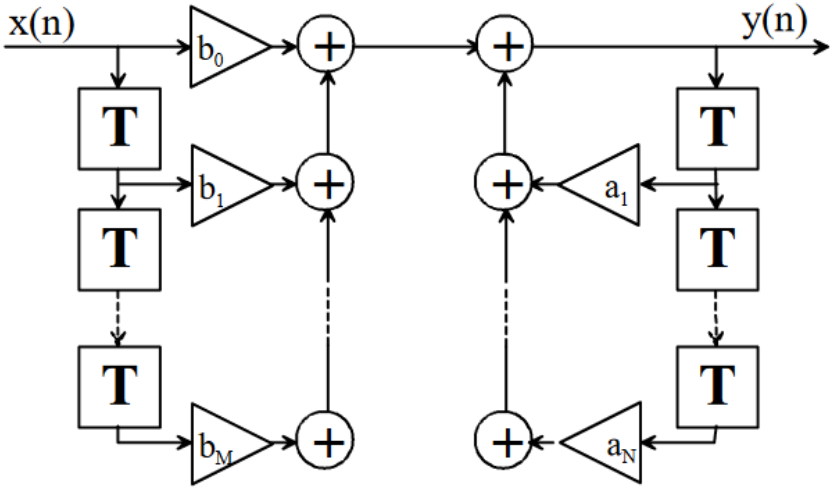


Figure 5: IIR

Question 20. A filter has the following difference equation:

$$y(n) + 0.2y(n-1) = 0.01y(n-2) + 0.3x(n) + 0.2x(n-1)$$

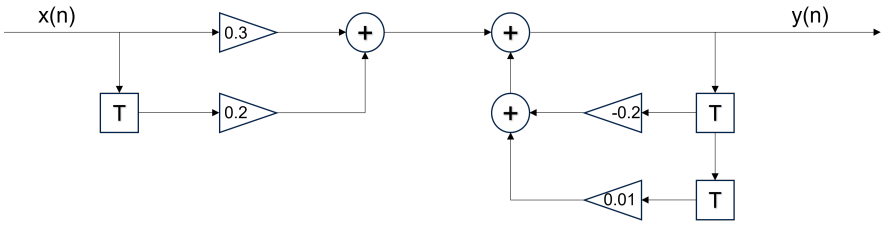
1. Draw the filter on a standard IIR filter format.

2. Compute the output $y(n)$ when the input is

$$x(n) = 0, n < 0, n > 4,$$

$x(n) = \{2, -3, 1, -1\}$, when $n = 1, 2, 3, 4$. Assume a causal system.

Answer:



1.

2. Computing the output y(n):
   Rearranging the difference equation:

$$y(n) = -0.2y(n-1) + 0.01y(n-2) + 0.3x(n) + 0.2x(n-1),$$

   and assuming $y(z) = 0$ for $z \leq 0$ we now get

$$y(0) = 0 \tag{18}$$
$$y(1) = 0.6 \tag{19}$$
$$y(2) = -0.62 \tag{20}$$
$$y(3) = -0.17 \tag{21}$$
$$y(4) = 0.0722 \tag{22}$$
$$y(5) = -0.18726 \tag{23}$$

   and for $k > 5$, we get $y(k) = -0.2y(k-1) + 0.01y(k-2)$, which converges to 0 for $k \mapsto \infty$.

Question 21.

1. What signifies a FIR and IIR filter, respectively? (cf. advantages and disadvantages)
2. Write the difference equation for a general FIR and IIR, respectively.
3. How many memory positions does an IIR filter require that have M forward taps and N iteration taps? (Are you sure? – Search the web!)

Answer:

1. Answers can be found on lecture 7 slide 24
   FIR:
   - +: Always stable
   - +: Enables linear phase response

- +: (Easy to design)
- -: Require higher orders (long delays and more computations)

IIR:

- +: Possible with sharper transitions
- +: Efficient implementations (less delay)
- -: Potential instability
- -: Phase distortion

2. Difference equations:

$$y_{\text{FIR}}(n) = \sum_{k=0}^{M} b_k x(n-k) \tag{24}$$

$$y_{\text{IIR}}(n) = \sum_{k=1}^{N} a_k y(n-k) + \sum_{k=0}^{M} b_k x(n-k) \tag{25}$$

3. Let us introduce intermediate states

$$w(n) = x(n) + \sum_{k=1}^{N} b_k w(n-k) \tag{26}$$

and the resulting output

$$y_{\text{IIR}}(n) = \sum_{k=0}^{M} a_k w(n-k), \tag{27}$$

leading to a memory requirement of $\max\{N, M\}$

Question 22.

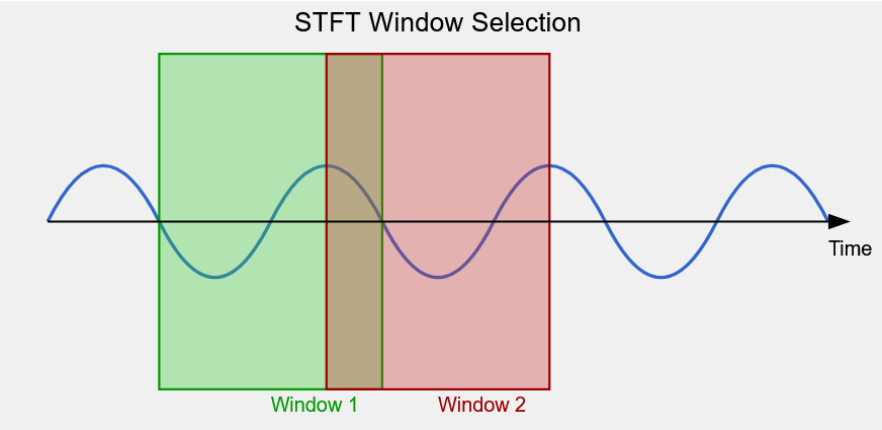
1. What is the purpose of the STFT? How is it used?
2. Write the formula for the Short-Time Fourier Transform.
3. Draw a figure that explains how the STFT window selects the time elements.
4. What is the trade-off when choosing the window function?

Answer:

1. The purpose of the Short-Time Fourier Transform (STFT) is to analyze how the frequency content of a signal changes over time. By employing a window function, it allows for frequency analysis of a signal at a specific point (duration) in time. It is used for time-frequency analysis of non-stationary signals.
2. STFT formula:

$$X(t, \omega) = \int_{-\infty}^{\infty} x(\tau) w(t-\tau) e^{-j\omega\tau} d\tau$$

where $w(t)$ is the window function.



3.

4. A large window allows for higher resolution in the frequency domain but a lower resolution in time domain

5. For a smaller window the opposite holds true

Question 23.

1. How is the Spectrogram different from the STFT?

2. Give examples of use of the STFT.

Answer:

1. The Spectrogram is a visual representation of the STFT. It displays the magnitude of the STFT as a 2D plot with time on one axis, frequency on the other, and the magnitude represented by color or intensity.

2. Examples of STFT:

   - Speech processing
   - Music analysis
   - Biomedical signal processing
   - Mechanical vibration analysis